DJBILBOX BEATS / HUMPIRE

K2000

Version 1.0 - Windows (VST3) and macOS (VST3 / Audio Unit) - Royalty-free license

Contents

K2000

- Overview
- Quick start in FL Studio
- Interface
- Parameters
- Factory presets
- Production tips
- MIDI, automation and performance
- FAQ

K2000

80s synthwave and darksynth synthesizer: leads, basses, pads, plucks, keys and brass, with a ladder filter, an LFO with its own envelope and a full effects chain.

Type	Synthesizer (instrument)
Format	VST3 64-bit (Windows); VST3 + Audio Unit, universal (macOS)
Brand / developer	HUMPIRE by DJBILBOX BEATS
Version	1.0
Factory presets	70
Latency	0 samples

Overview

K2000 is an 80s-style synthesizer sold under the HUMPIRE brand by DJBILBOX BEATS. It uses no samples. Every sound is generated in real time by one oscillator section with six waveforms (Sawtooth, Square, Triangle, Sine, SuperSaw and 80s Brass), followed by a multimode filter, an amplitude envelope and an LFO that has its own ADSR envelope.

The sound then goes through a master effects chain: distortion, a stereo chorus, a ping-pong tape-style delay and a reverb, and finally a volume stage with a soft limiter. The look is a neon-cyan faceplate with a cassette-style EXPANSION slot, pitch and modulation wheels and an on-screen keyboard.

K2000 targets Synthwave, Darksynth, Retrowave, Synthpop, Outrun and 80s-style soundtrack work: lead lines, bass lines, wide pads, plucks and arps, keys, brass stabs and atmospheric effects. The filter offers five modes, including a 24 dB ladder and an “Acid” low pass. Voice mode is selectable: Poly 8, Poly 16, Mono Retrig and Mono Legato. Polyphony is up to 16 voices.

Quick start in FL Studio

1. In the **Channel rack**, click **+** and choose **K2000** (vendor **DJBILBOX BEATS**). You can also drag it from the **Plugin database > Generators** browser.
2. The plugin opens on **K2000 - Init Patch**: a Sawtooth waveform in Poly 8 mode.
3. Click the **folder icon** at the left of the preset display. Pick a bank, then a preset. You can also step through all 70 presets with the **<** and **>** arrows.
4. Play from your MIDI keyboard or draw notes in the **Piano roll**. The on-screen keyboard covers MIDI notes 36 to 84, which is C3 to C7 in FL Studio naming (where middle C is C5).
5. Shape the sound with the main controls:
   - **WAVEFORM** menu: the raw character of the oscillator.
   - **CUTOFF**: the filter cutoff. Turn it down for a darker, rounder sound.
   - **RES**: the filter resonance, for a sharper, more vocal sweep.
   - **GLIDE**: slide time between notes. Turn it down for straight notes.
6. Adjust **VOLUME** (right side of the panel) to set the output level, then click the **disk icon** to save your own version.

Interface

The window opens at 1000 x 746 pixels. Drag the bottom-right corner to resize it from 50% to 200%. The proportions stay fixed.

Preset display (top center). From left to right: the folder icon (preset menu), the **<** arrow, the current preset name, the **>** arrow and the disk icon (save). The arrows wrap around from the last preset to the first.

EXPANSION cassette (top left). Clicking the cassette opens an information message announcing that expansion packs are coming soon. No expansion content is included with version 1.0.

Logo (top right). The DJBILBOXBEATS name and the DB logo.

Center. The camel artwork, the K2000 title and “By Humpire”.

Left vent panel (beside the camel). Two small knobs: **RES** (filter resonance) and **FLT ENV** (filter envelope depth).

Right vent panel (beside the camel). Two small knobs: **AMP REL** (amplitude release time) and **DETUNE** (spread of the SuperSaw waveform).

AMP ENVELOPE and VELOCITY (top right). Five vertical sliders: **ATTACK, DECAY, SUSTAIN** (the amplitude envelope) and **BANK, ADSR** (the two velocity controls, under the VELOCITY header). The amplitude release is the **AMP REL** knob in the right vent panel.

LFO section (left). RATE and **AMOUNT** set the LFO speed and depth. The **LFO>PITCH** switch between them sends a light vibrato to the pitch, whatever the LFO DESTINATION menu says. To the right, the **LFO ENVELOPE** has four sliders: **ATTACK, DECAY, SUSTAIN, RELEASE**. They shape how the LFO depth fades in and out during each note.

Center and right knobs.

- Row 1, right: **VOLUME, REVERB, CHORUS, DISTORTION**. DISTORTION is the master distortion on the summed output.
- Row 2, left to right: **GLIDE, CUTOFF** (center), then **DRIVE** (filter input drive), **DLY MIX** (delay level), **DLY TIME** (delay time) and **FEEDBACK** (delay repeats).

Menu row. Five drop-down menus, each with a caption below it: **WAVEFORM**, **LFO SHAPE**, **LFO DESTINATION**, **FILTER TYPE** and **VOICE MODE**. Each menu shows its current value.

Wheels (bottom left). PITCH BEND springs back to the center when you release it. **MODULATION** stays where you leave it.

On-screen keyboard (bottom). Click a key to play it. Velocity follows the click position: click near the bottom of a key for a louder note. The keyboard covers MIDI notes 36 to 84 (four octaves plus one key). The key labels on screen run from C1 to C5, with middle C labeled C3; in FL Studio naming, where middle C is C5, the same range is C3 to C7. It does not scroll. For notes outside this range, use a MIDI controller or the Piano roll.

Mouse gestures. Drag a knob up or right to increase it. Drag sliders and wheels vertically. A value bubble appears while you drag. You can also type a value, for example `440 Hz` or `35`, in the host’s parameter field.

Parameters

Signal flow: oscillator, then filter, then amplitude envelope (per voice, with the LFO and the filter envelope acting on pitch, cutoff, pulse width, volume and pan). The voices are summed, then processed by Distortion, Chorus, Delay and Reverb. A DC blocker and a soft limiter close the chain, after the VOLUME control.

K2000 exposes 33 automatable parameters. The host name is shown in parentheses when it differs from the panel label.

Parameter	Range	Default	Function
WAVEFORM (Waveform menu)	Sawtooth, Square, Triangle, Sine, SuperSaw, 80s Brass	Sawtooth	Selects the oscillator waveform. Square is a pulse wave with a 50% duty cycle. SuperSaw stacks seven detuned saws. 80s Brass blends a saw, a pulse wave and an octave-up sine.
DETUNE (Osc Detune)	0 to 100%	40%	Spread of the seven SuperSaw oscillators (up to about +/-2% in frequency at 100%). On 80s Brass it adds a very small detune (up to 0.3%). No effect on the other waveforms.
CUTOFF (Filter)	20 to 20,000 Hz	3,200 Hz	Filter cutoff. The cutoff is not keyboard-tracked.
RES (Filter Resonance)	0 to 100%	20%	Filter resonance.
FILTER TYPE (menu)	LowPass 24dB, LowPass 12dB, BandPass, HighPass, Acid LowPass	LowPass 24dB	LowPass 24dB is a 4-pole ladder filter. LowPass 12dB, BandPass and HighPass use a 12 dB state-variable filter. Acid LowPass is the ladder with extra resonance and drive.
DRIVE (Distortion Drive)	0 to 100%	15%	Drive applied at the filter input (per voice). Not the master distortion.
FLT ENV (Filter Env Amount)	0 to 100%	0%	Depth of the filter envelope. At 100% the envelope opens the cutoff by up to 5 octaves. The envelope is fixed: 2 ms attack, decay equal to the amplitude DECAY (limited to 0.03 to 3 s), sustain 0, release 0.1 s.
ATTACK (Amp Attack)	1 ms to 5 s	20 ms	Amplitude envelope attack time.
DECAY (Amp Decay)	5 ms to 8 s	0.8 s	Amplitude envelope decay time. It also sets the filter envelope decay.
SUSTAIN (Amp Sustain)	0 to 100%	70%	Amplitude envelope sustain level.
AMP REL (Amp Release)	10 ms to 8 s	0.4 s	Amplitude envelope release time.
BANK (Velocity Bank)	0 to 100%	75%	Velocity depth on the note level. At 0% all notes have the same level; at 100% the level follows the velocity fully.

Parameter	Range	Default	Function
ADSR (Velocity ADSR)	0 to 100%	40%	Additional velocity response on the amplitude envelope (a curved response, so soft notes get quieter). At 0% it has no effect.
GLIDE (Glide)	0 to 1 (time scale in seconds)	0.05	Portamento. Each new note slides from the pitch of the previous note, in every voice mode. Values below about 0.002 turn glide off.
RATE (LFO Rate)	0.05 to 25 Hz	4.5 Hz	LFO speed. Not tempo-synced.
AMOUNT (LFO Amount)	0 to 100%	35%	LFO depth. The mod wheel adds to this value.
LFO SHAPE (LFO Source menu)	Sine, Triangle, Square, Saw Up, Saw Down, Random S&H	Triangle	LFO shape. The LFO restarts at every new note. Random S&H picks a new random value once per cycle.
LFO DESTINATION (Destination menu)	Cutoff, Pitch, Pulse Width, Volume, Chorus, Pan	Cutoff	LFO target. Cutoff: up to +/-4 octaves. Pitch: up to +/-1 semitone. Pulse Width: pulse width from 5% to 95% (audible on Square and, in part, on 80s Brass). Volume: up to a 50% level dip (tremolo). Chorus: moves the chorus depth by up to +/-0.5. Pan: moves each voice across the stereo field.
LFO>PITCH (LFO Pitch Switch)	Off, On	Off	When on, adds a vibrato of up to +/-0.5 semitone from the LFO, in addition to the selected destination.
LFO ENVELOPE ATTACK (LFO Attack)	1 ms to 5 s	0.1 s	Time for the LFO depth to fade in at each note.
LFO ENVELOPE DECAY (LFO Decay)	5 ms to 8 s	0.5 s	Decay time of the LFO envelope.
LFO ENVELOPE SUSTAIN (LFO Sustain)	0 to 100%	100%	Sustain level of the LFO depth. At 0% the LFO fades out after the attack and decay.
LFO ENVELOPE RELEASE (LFO Release)	5 ms to 8 s	0.4 s	Release time of the LFO envelope.
VOICE MODE (Mono/Poly menu)	Poly 8, Poly 16, Mono Retrig, Mono Legato	Poly 8	Poly 8: up to 8 notes at once. Poly 16: up to 16 notes. Mono Retrig: one note at a time, envelopes restart at every note. Mono Legato: one note at a time, envelopes are not restarted when you play legato.

Parameter	Range	Default	Function
DISTORTION (Distortion Master)	0 to 100%	25%	Saturation on the summed output, with level compensation.
CHORUS (Chorus)	0 to 100%	45%	Stereo chorus depth.
DLY MIX (Delay Mix)	0 to 100%	30%	Level of the ping-pong delay.
DLY TIME (Delay Time)	10 ms to 1 s	0.375 s	Delay time. Not tempo-synced: for a quarter note use 60 / BPM (0.5 s at 120 BPM). The right channel is 15% longer than the left.
FEEDBACK (Delay Feedback)	0 to 95%	45%	Number of delay repeats. The repeats are darkened and slightly saturated.
REVERB (Reverb)	0 to 100%	35%	Reverb amount.
VOLUME (Volume)	0 to 100% (-inf to 0 dB)	80% (-1.9 dB)	Master output level, applied before the soft limiter.
Pitch Bend wheel (Pitch Bend)	-100 to +100% (-200 to +200 cents)	0	On-screen pitch bend, +/-2 semitones. Adds to MIDI pitch bend.
Modulation wheel (Mod Wheel)	0 to 100%	0	On-screen mod wheel. Adds to the LFO AMOUNT, opens the cutoff by up to 0.8 octave and raises the chorus depth by up to 25%.

Factory presets

K2000 includes 70 factory presets in 7 banks. Voice modes across the 70 presets: 43 load in Poly 8, 3 in Poly 16, 10 in Mono Retrig and 14 in Mono Legato.

VOL.1 Factory (17): K2000 - Init Patch, 01 - Neon Night Drive, 02 - Outrun Bassline, 03 - Blade Runner Lead, 04 - Miami Vice Pad, 05 - Turbo Arp Pluck, 06 - VHS Tape Dreams, 07 - Cyber Chaser Bass, 08 - Sunset Grid Pluck, 09 - Stranger Horror Pad, 10 - Megadrive Solo Lead, 11 - Chrome Overdrive, 12 - Digital Rain Pluck, 13 - Retrowave Brass Stabs, 14 - Cyborg Camel Cry, 15 - Cassette LoFi Keys, 16 - Future Hero Solo.

Leads (9): 17 - Laser Grid Lead, 18 - Sunset Strip Lead, 19 - Neon Sirens, 20 - Turbo Brass Lead, 21 - Pulse Rider, 22 - Crystal Hero, 23 - Acid Rain Lead, 24 - Lost Highway, 25 - Cyber Whistle.

Bass (9): 26 - Night Pulse Bass, 27 - Sub Highway, 28 - Darksynth Growl, 29 - Octave Chase, 30 - Moog Miami, 31 - Rubber Grid, 32 - Terminator Sub, 33 - Wobble Chrome, 34 - Hi-NRG Bass.

Pads (9): 35 - Sunset Strip Pad, 36 - Blade Fog, 37 - Outrun Strings, 38 - Neon Choir, 39 - Dark Cinema, 40 - Aurora Slow Pad, 41 - Cassette Warmth, 42 - Retro Glass Pad, 43 - Cyborg Breath.

Plucks & Arps (9): 44 - Arp Machine, 45 - Neon Pluck, 46 - Glass Arp, 47 - Sequence Chrome, 48 - Dream Bell, 49 - Laser Tag, 50 - Night Marimba, 51 - Filter Gate, 52 - Echo Pluck.

Keys & Brass (9): 53 - Retro Electric Keys, 54 - Synthpop Keys, 55 - Power Brass, 56 - 80s Horns, 57 - Stab Section, 58 - Soft Strings Keys, 59 - Chip Organ, 60 - Polysynth Chord, 61 - Jazz Fusion Keys.

FX & Atmos (8): 62 - Warp Drive, 63 - Nuclear Dawn, 64 - Sonar Ping, 65 - Static Rain, 66 - Tape Stop Dream, 67 - Hologram, 68 - Overheating, 69 - Last Exit.

Loading a preset first resets every parameter to its default value, then applies the preset. The two wheels are not changed.

User presets. Click the disk icon to save the current sound as a `.k2preset` file. The default folder is `DJBILBOX BEATS/K2000/Presets` inside your Documents folder. Files in this folder appear in the **User** submenu of the preset menu. Use **Open a .k2preset file…** in the same menu to load a file from another location. The preset display shows the file name. User presets do not change the two wheels.

The host also sees the 70 factory presets as a program list. Your project saves the full state of the plugin, including the preset name.

“Moog Miami” is a preset name that refers to a sound style. Moog is a trademark of its owner. DJBILBOX BEATS is not affiliated with the trademark owner.

Production tips

1. **Synthwave leads.** Start from Laser Grid Lead or Sunset Strip Lead (Mono Legato). Raise GLIDE slightly for slides between notes, and keep the LFO>PITCH switch on for a light vibrato. Add DLY MIX and REVERB for the classic wide lead.
2. **Outrun and darksynth bass.** Use the Bass bank (Night Pulse Bass, Terminator Sub, Darksynth Growl). Raise FLT ENV and shorten the amplitude DECAY for a punchy filter pluck on every note. Raise RES for more bite.
3. **Wobble.** Set LFO DESTINATION to Cutoff, LFO SHAPE to Sine or Triangle and RATE to 2 Hz or so, as in Wobble Chrome. Keep AMOUNT above 50% for a deep movement.
4. **Wide pads.** The Pads bank uses long attacks. For extra width use the SuperSaw waveform, raise DETUNE and CHORUS, and add REVERB.
5. **Arps and plucks.** The Plucks & Arps bank uses short decays and zero sustain. Feed it a fast arpeggio from your Piano roll or a MIDI arpeggiator, then set DLY TIME to a musical value with moderate FEEDBACK. At 120 BPM, 0.375 s is a dotted eighth note (3 x 60 / BPM / 4).
6. **Brass stabs.** Choose the 80s Brass waveform, a medium ATTACK and a low SUSTAIN (see Stab Section and Retrowave Brass Stabs). Add a little CHORUS.
7. **Pulse width.** Select the Square waveform, set LFO DESTINATION to Pulse Width and RATE to a slow value for the classic moving pulse sound.
8. **Stereo movement.** Set LFO DESTINATION to Pan with a Triangle or Sine LFO and a low RATE for a slow auto-pan across the voices.
9. **Level and dynamics.** BANK sets how much velocity changes the level. Set it near 75% for natural accents, or 0% for even notes.

MIDI, automation and performance

Notes and channels. K2000 responds to all 128 MIDI notes on any MIDI channel. Velocity controls the level according to the BANK and ADSR sliders.

Polyphony. Up to 16 voices in Poly 16 and up to 8 voices in Poly 8. When you play more notes than the limit, a voice is taken over and cut. In both Mono modes, each new note stops the previous one (last-note priority). Releasing the newest note while an earlier note is still held brings that earlier note back.

Pitch bend. MIDI pitch bend covers +/-2 semitones. The on-screen PITCH BEND wheel adds another +/-2 semitones.

Mod wheel (CC1). CC1 raises the LFO depth and opens the filter (up to 0.8 octave), like the on-screen MODULATION wheel. When both are used on a voice, the higher value wins for the LFO and filter. The chorus depth increase follows the on-screen **Mod Wheel** parameter only. For repeatable results in a song, automate the **Mod Wheel** parameter.

Other MIDI. The sustain pedal (CC64) is supported in the Poly modes and is ignored in the Mono modes. All Notes Off, All Sound Off and Reset All Controllers messages are handled. Aftertouch is ignored.

Automation in FL Studio. All 33 parameters can be automated. Move a control in the plugin, then use **Tools > Last tweaked > Create automation clip**. Use **Tools > Last tweaked > Link to controller** to assign a hardware knob. You can also pick parameters from the plugin wrapper menu (**Browse parameters**).

Latency and tail. The plugin adds no latency (0 samples), so no delay compensation is needed. It reports a 12-second tail to the host so reverb and delay trails are not cut when rendering.

CPU. The master DISTORTION is bypassed when set to zero. Poly 8 and short release times keep the voice count low. SuperSaw uses seven oscillators per voice, so it costs more than the other waveforms. The plugin was tested at 44.1, 48, 88.2 and 96 kHz.

FAQ

K2000 does not appear in FL Studio. Open **Options > Manage plugins**, check that `C:\Program Files\Common Files\VST3` is in the search paths, then click **Find installed plugins**. The plugin is listed as **K2000** under **DJBILBOX BEATS**. It is an instrument: load it in the Channel rack, not in a Mixer insert slot.

I hear nothing when I click the on-screen keyboard. Check VOLUME, and check that CUTOFF is not turned all the way down with FILTER TYPE set to a low pass mode. Some presets (Bass bank, Pads) can sound soft on small speakers. For notes outside the on-screen keyboard range (MIDI 36 to 84), play from a MIDI keyboard or the Piano roll.

Every note slides into the next one. GLIDE is above zero. Glide applies to every new note, in every voice mode. Set GLIDE to 0, or automate it only where you want slides.

My chord only plays one or two notes. The VOICE MODE menu is set to Mono Retrig or Mono Legato (or Poly 8 with more than 8 notes). Switch it to Poly 8 or Poly 16.

The filter envelope does nothing. FLT ENV is 0% by default. Raise it to hear the filter opening at each note. The envelope's decay follows the amplitude DECAY slider.

Does it work without a MIDI keyboard? Yes. Draw notes in the Piano roll or click the on-screen keyboard.

Where do I install it on macOS? Copy the VST3 to `/Library/Audio/Plug-Ins/VST3` and the Audio Unit to `/Library/Audio/Plug-Ins/Components`, then rescan plugins in your host. Both are universal builds.